

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VII (NEW) EXAMINATION – WINTER 2023****Subject Code:3174205****Date:08-12-2023****Subject Name: Natural Language Processing****Time: 10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Marks

- | | | |
|------------|--|-----------|
| Q.1 | (a) State advantages of Natural Language Processing. | 03 |
| | (b) Explain applications of language modeling. | 04 |
| | (c) Discuss phases of Natural Language Processing with suitable example. | 07 |
| Q.2 | (a) State applications of language modeling. | 03 |
| | (b) Write a short note on named entity recognition. | 04 |
| | (c) Explain Unigram, Bi-gram and Tri-gram concepts using an example sentence: "This is a sentence". | 07 |
| OR | | |
| | (c) Explain various smoothing techniques with example. | 07 |
| Q.3 | (a) Define one sense per discourse with example. | 03 |
| | (b) Explain structured polysemy. | 04 |
| | (c) Suppose you are doing bag-of-words text classification on a document. The raw input is a single string containing the text of the entire document. Describe the pipeline to go from the raw input to a feature vector. | 07 |
| OR | | |
| Q.3 | (a) Define following terms: gloss, synset, supersense | 03 |
| | (b) Explain IS-A hierarchy. | 04 |
| | (c) Describe the reason why skip-gram with negative sampling is faster to train than the original skip-gram model. | 07 |
| Q.4 | (a) Give attributes of text classification. | 03 |
| | (b) Write a short note on Relation Extraction via Supervised Learning. | 04 |
| | (c) Analyze any application of NLP using the concept of Cross-Lingual IR. | 07 |
| OR | | |
| Q.4 | (a) Give the advantages of sentiment mining. | 03 |
| | (b) Explain information extraction procedure. | 04 |
| | (c) Explain the role of confusion matrix. Create a confusion matrix for visualizing how well a binary classification system performs against gold standard labels. | 07 |
| Q.5 | (a) Define the need of Machine Translation. | 03 |
| | (b) Explain objective of rule-based Machine Translation. | 04 |
| | (c) Write a short note on Neural-Machine Translation. | 07 |
| OR | | |
| Q.5 | (a) List out problems of Machine Translation. | 03 |
| | (b) Draw the basic architecture of Encoder-Decoder model in Machine Translation. | 04 |
| | (c) Write a short note on Statistical Machine Translation. | 07 |
